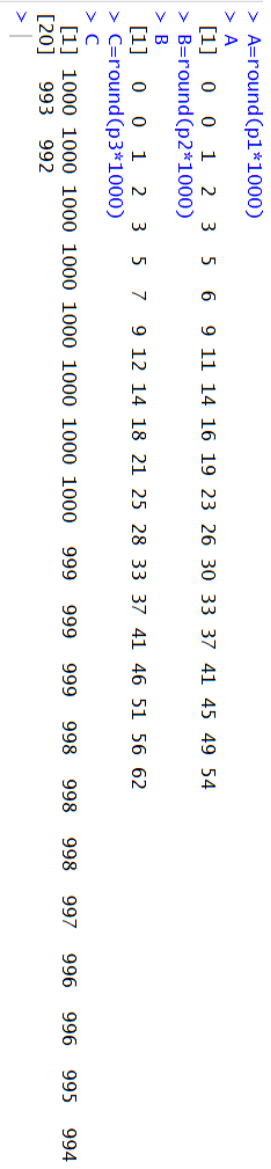


The graph illustrates the Poisson distribution, which models the probability of a given number of events occurring in a fixed interval of time or space. The x-axis represents the number of random values (0 to 20), and the y-axis represents the probability values (0.0 to 1.0). The distribution is highly skewed to the right, with the highest probability at x=0 (approximately 0.99) and a rapid decay towards zero as x increases.

random values	Probability values
0	0.99
1	0.01
2	0.00
3	0.00
4	0.00
5	0.00
6	0.00
7	0.00
8	0.00
9	0.00
10	0.00
11	0.00
12	0.00
13	0.00
14	0.00
15	0.00
16	0.00
17	0.00
18	0.00
19	0.00
20	0.00



```
> A=round(p1*1000)
> A
[1] 0 0 1 2 3 5 6 9 11 14 16 19 23 26 30 33 37 41 45 49 54
> B=round(p2*1000)
> B
[1] 0 0 1 2 3 5 7 9 12 14 18 21 25 28 33 37 41 46 51 56 62
> C=round(p3*1000)
> C
[1] 1000 1000 1000 1000 1000 1000 1000 1000 999 999 999 998 998 998 997 996 996 995 994
[20] 993 992
```

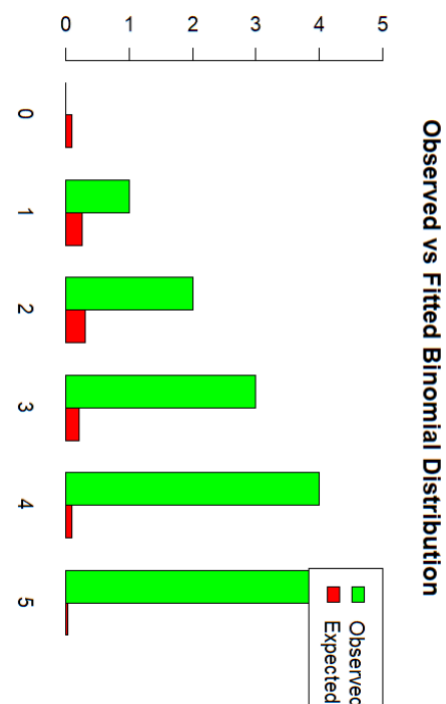
#A manufacturer of pins knows that 2% of his products are defective. If he calls pins in boxes of 10, what is the probability that there will be 2 defective pins in boxes of 10?

```
> x=0:5
> f=c(2,5,6,4,2,1)
> f
[1] 2 5 6 4 2 1
> #mean
> N=sum(f)
> mean=sum(x*f)/N
> mean
[1] 2.1
> n=10
> p=mean/n
> p
[1] 0.21
> q=1-p
> prob=dbinom(x,n,p)
> prob
[1] 0.09468276 0.25168835 0.30107024 0.21341688 0.09927937 0.03166886
> expected=N*prob
> expected
[1] 1.8936552 5.0337670 6.0214049 4.2683376 1.9855874 0.6333773
> #create table
> result=data.frame(x=x,f=prob,Observed=round(prob,3),Expected=round(expected,2))
> result
  x      f Observed Expected
1 0 0.09468276   0.095    1.89
2 1 0.25168835   0.252    5.03
3 2 0.30107024   0.301    6.02
4 3 0.21341688   0.213    4.27
5 4 0.09927937   0.099    1.99
6 5 0.03166886   0.032    0.63
  main="Poisson Distribution".
```

#The following data represents the number of printing errors found on 100 pages of a books

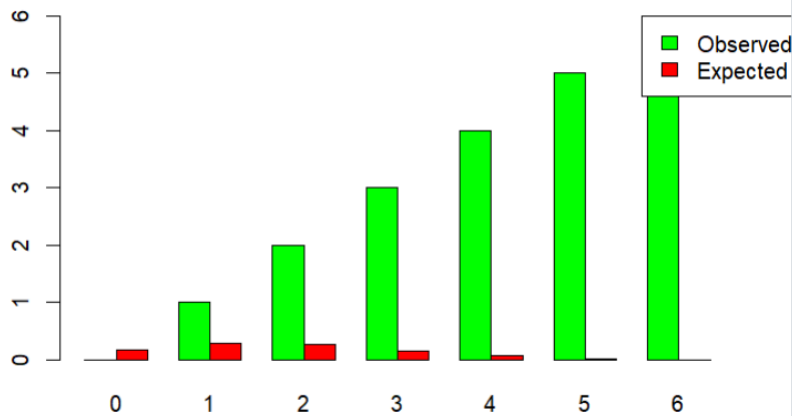
```
#No. of errors per page(x) 0 1 2 3 4 5 6
#Frequency(f)             20 28 24 15 8 4 1
#fit a poisson distribution to the above data and find the expected freq
```

```
x=0:6
f=c(20,28,24,15,8,4,1)
f
N=sum(f)
N
mean=sum(x*f)/N
mean
lambda=mean
lambda
prob=dpois(x,lambda)
prob
expected=N*prob
expected
result=data.frame(x=x,f=prob,Observed=round(prob,3),Expected=round(expected,2))
result
barplot(rbind(x,prob),
        beside=TRUE,
        names.arg=x,
        col=c("green","red"),
        legend.text=c("Observed","Expected"),
        main="Observed vs Fitted Poisson Distribution",
        args.legend=list(x="topright",inset=c(-0.1,0)))
```



```
> f
[1] 20 28 24 15 8 4 1
> N=sum(f)
> N
[1] 100
> mean=sum(x*f)/N
> mean
[1] 1.79
> lambda=mean
> lambda
[1] 1.79
> prob=dpois(x,lambda)
> prob
[1] 0.166960170 0.298858704 0.267478540 0.159595529 0.071418999 0.025568002 0.007627787
> expected=N*prob
> expected
[1] 16.6960170 29.8858704 26.7478540 15.9595529 7.1418999 2.5568002 0.7627787
> result=data.frame(x=x,f=prob,Observed=round(prob,3),Expected=round(expected,2))
> result
  x      f Observed Expected
1 0 0.166960170   0.167   16.70
2 1 0.298858704   0.299   29.89
3 2 0.267478540   0.267   26.75
4 3 0.159595529   0.160   15.96
5 4 0.071418999   0.071    7.14
6 5 0.025568002   0.026    2.56
7 6 0.007627787   0.008    0.76
```

Observed vs Fitted Poisson Distribution



```
#The number of defective items found in 20 randomly selected batches where each batch contains 10 items
#No. of defectives(x):0 1 2 3 4 5
#Frequency(f): 2 5 6 4 2 1
#Fit a binomial Distribution to the above data and determine the expected frequencies for each possible
#number of defectives.
```

```
x=0:5
f=c(2,5,6,4,2,1)
#mean
N=sum(f)
mean=sum(x*f)/N
mean
n=10
p=mean/n
q=1-p
prob=dbinom(x,n,p)
prob
expected=N*prob
expected
#create table
result=data.frame(x=x,f=prob,Observed=round(prob,3),Expected=round(expected,2))
result
#Visualize
barplot(rbind(x,prob),
        beside=TRUE,
        names.arg=x,
        col=c("green","red"),
        legend.text=c("Observed","Expected"),
        main="Observed vs Fitted Binomial Distribution",
        args.legend=list(x="topright",inset=c(-0.1,0)))
```

```
> #a company finds that the time taken by one of its engineers to complete or repair job has a normal distribution
> #with mean 20 minutes and sd 5 minutes find prob of i. less than 15 ii. more than 25 iii. 15-25 iv. plot v.table
>
> x = 0:40
> m = 20
> s = 5
> p1 = pnorm(15,m,s)
> p1#less than 15
[1] 0.1586553
> p2 = 1-pnorm(25,m,s)
> p2#more than 25
[1] 0.1586553
> p3 = pnorm(25,m,s) - pnorm(15,m,s)
> p3# 15-25
[1] 0.6826895
> y = dnorm(x,m,s)
> y
[1] 2.676605e-05 5.838939e-05 1.223804e-04 2.464438e-04 4.768176e-04 8.863697e-04 1.583090e-03 2.716594e-03
[9] 4.478906e-03 7.094919e-03 1.079819e-02 1.579003e-02 2.218417e-02 2.994549e-02 3.883721e-02 4.839414e-02
[17] 5.793831e-02 6.664492e-02 7.365403e-02 7.820854e-02 7.978846e-02 7.820854e-02 7.365403e-02 6.664492e-02
[25] 5.793831e-02 4.839414e-02 3.883721e-02 2.994549e-02 2.218417e-02 1.579003e-02 1.079819e-02 7.094919e-03
[33] 4.478906e-03 2.716594e-03 1.583090e-03 8.863697e-04 4.768176e-04 2.464438e-04 1.223804e-04 5.838939e-05
[41] 2.676605e-05
> plot(x,y,type='l',xlab='time taken',ylab='density',main='normal distribution')
> x1 = seq(0,15)
> y1 = dnorm(x1,m,s)
> polygon(c(0,x1,15),c(0,y1,0),col='yellow')
> x2 = seq(25,40)
> y2 = dnorm(x2,m,s)
> polygon(c(25,x2,40),c(0,y2,0),col='green')
> x3 = seq(15,25)
> y3 = dnorm(x3,m,s)
> polygon(c(15,x3,25),c(0,y3,0),col='blue')
> data.frame(p1,p2,p3)
  p1      p2      p3
1 0.1586553 0.1586553 0.6826895
```

